DO NOW — RECALL

# Lesson 137 · Third Normal Form: Removing Transitive D

Name: _____ Class: _____ Date: _____

**1. What is a partial functional dependency?**

**2. What does ROM stand for?**

**3. What are the outputs of a half adder when A=1 and B=1?**

**4. What is a composite key?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

A non-key attribute depends on only part of a composite candidate key.

**2. Model answer**

Read-only memory.

**3. Model answer**

Sum 0, carry 1.

**4. Model answer**

A key made from two or more attributes together.

**By the end, you can…**

• Remove repeating groups and partial dependencies.

• EmployeeID → DeptID → DeptName; the name depends on department.

• Create Department(DeptID,DeptName) linked from Employee.

**Vocabulary — know these terms**

3NF — No transitive dependency of non-prime attributes on candidate keys.

transitive dependency — An attribute depends on a key indirectly through another non-key attribute.

functional dependency — Value of one attribute determines another.

update anomaly — Duplicate facts can disagree after an edit.

NEW LEARNING

# Read, annotate and explain

### Check 2NF first

Remove repeating groups and partial dependencies.

### Detect indirect dependency

EmployeeID → DeptID → DeptName; the name depends on department.

### Decompose tables

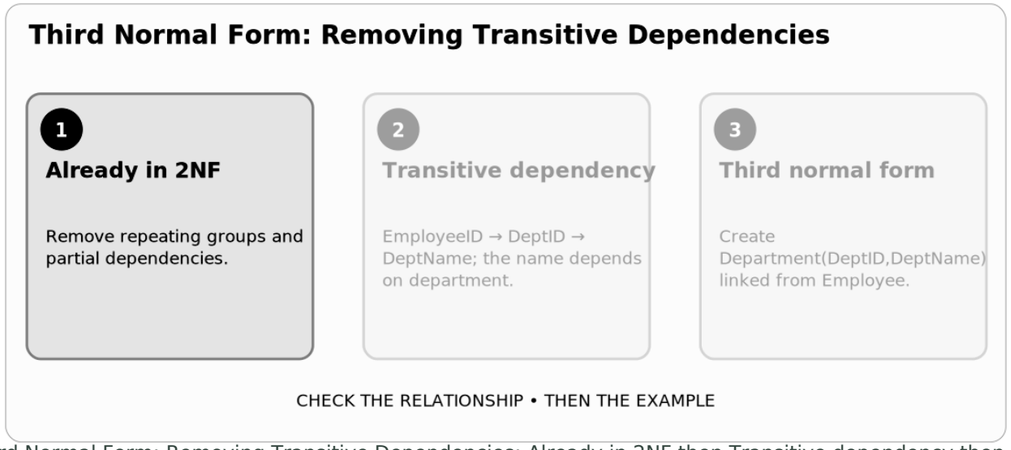
Create Department(DeptID,DeptName) linked from Employee.

**Explain one step in your own words, using the correct technical terms.**

# New Learning · original lesson diagrams

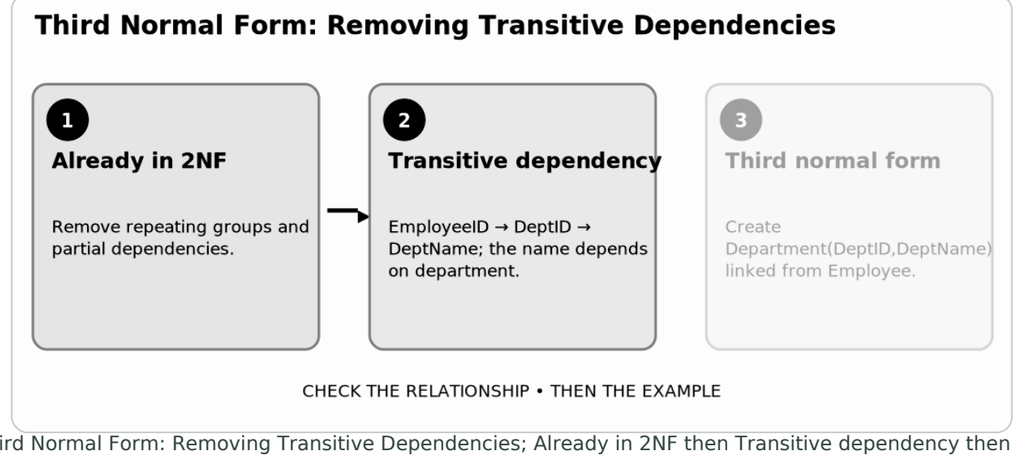
Diagrams appear after the related explanation. Annotate the original source visual.

### New Learning




Third Normal Form: Removing Transitive Dependencies; Already in 2NF then Transitive dependency then Third normal form

### New Learning




Third Normal Form: Removing Transitive Dependencies; Already in 2NF then Transitive dependency then Third normal form





# WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

### Guided 1 — original CHECK

Customer identifies postcode

Explain why this intermediate stage is necessary.

Split or preserve snapshot

Under the assumption CustomerID determines the customer's current postcode, there is an indirect dependency OrderID→CustomerID→CustomerPostcode. Store Customer(CustomerID PK,Postcode) and Order(OrderID PK,CustomerID FK). Historical shipping addresses may need separate, deliberate snapshots.

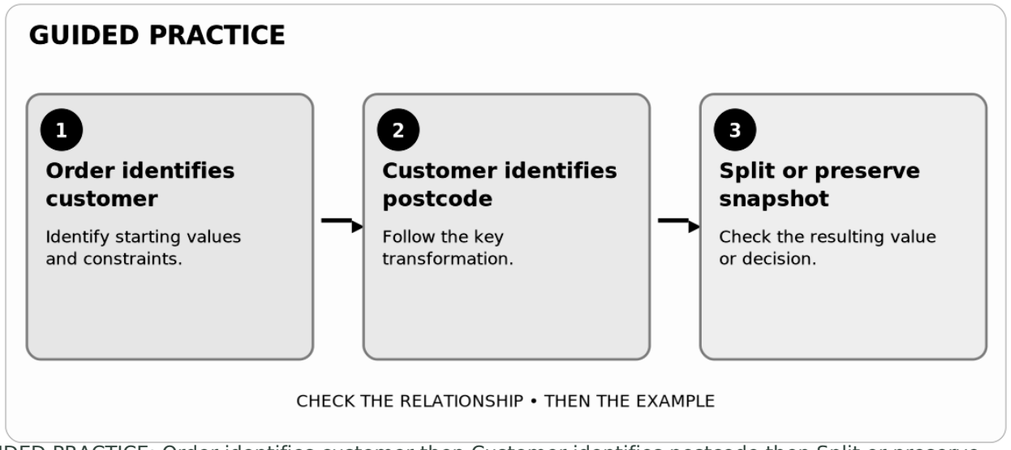
IMPROVE: Explain one correction to your own reasoning.

# We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

**Order(OrderID,CustomerID,CustomerPostcode) stores a customer's postcode on every order. Is a split justified?**

Find the given data and name the required result.




GUIDED PRACTICE; Order identifies customer then Customer identifies postcode then Split or preserve snapshot

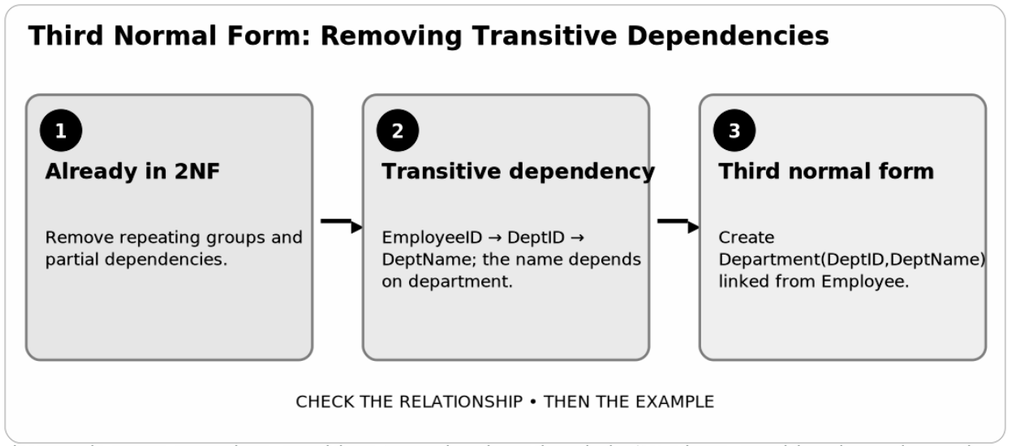




# New Learning · original lesson diagrams

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**New Learning**




Third Normal Form: Removing Transitive Dependencies; Already in 2NF then Transitive dependency then Third normal form

I DO — WORKED EXAMPLE

# Follow the worked method

**Employee(EmpID,EmpName,DeptID,DeptName) is in 2NF. Why may it violate 3NF?**

**EmpID determines DeptID**

Find the given data and name the required result.

**DeptID determines name**

Explain why this intermediate stage is necessary.

**Split Department**

EmpID identifies employee. DeptName depends on DeptID, not directly on EmpID, producing a transitive dependency. Split Employee(EmpID PK,EmpName,DeptID FK) and Department(DeptID PK,DeptName).



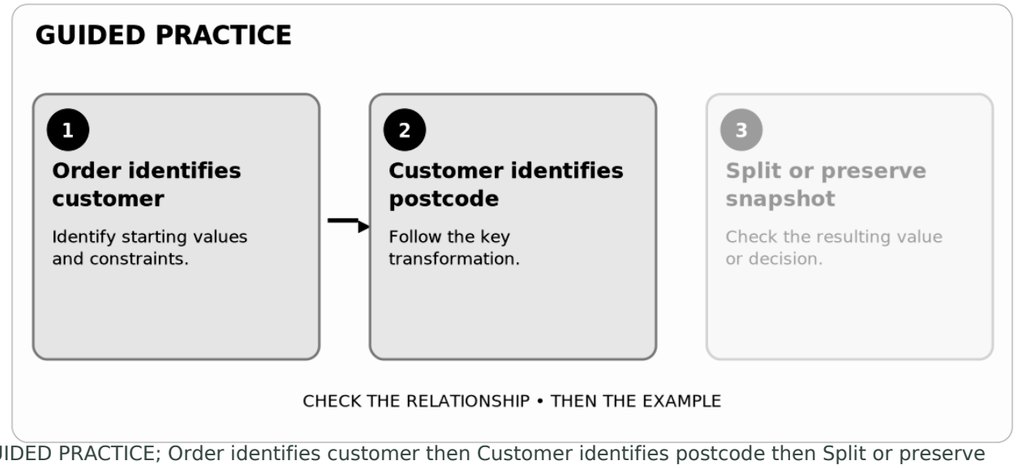


# We Do Check · original lesson diagrams

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GUIDED PRACTICE; Order identifies customer then Customer identifies postcode then Split or preserve snapshot

EXIT TICKET / REFLECTION

# Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**In 3NF, a non-key value should be copied into every child row merely for convenience.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.



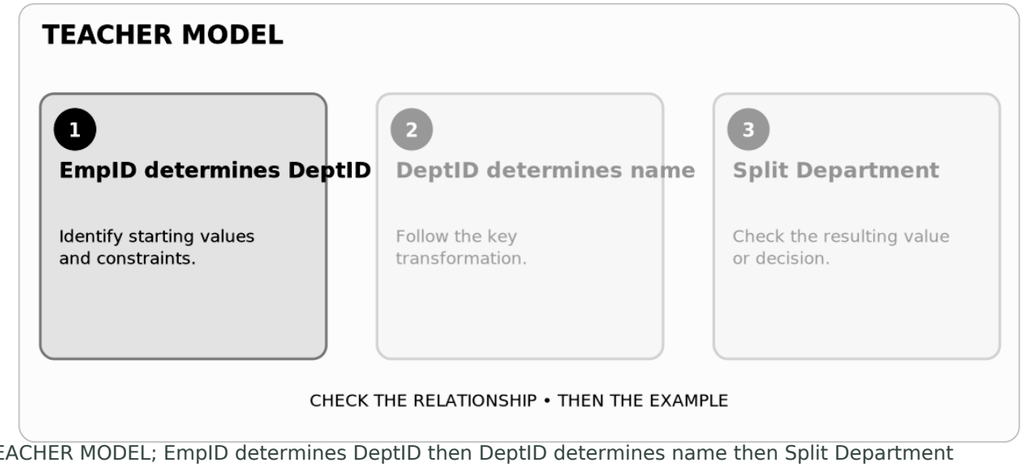


# I Do · original lesson diagrams

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**Employee(EmpID,EmpName,DeptID,DeptName) is in 2NF. Why may it violate 3NF?**

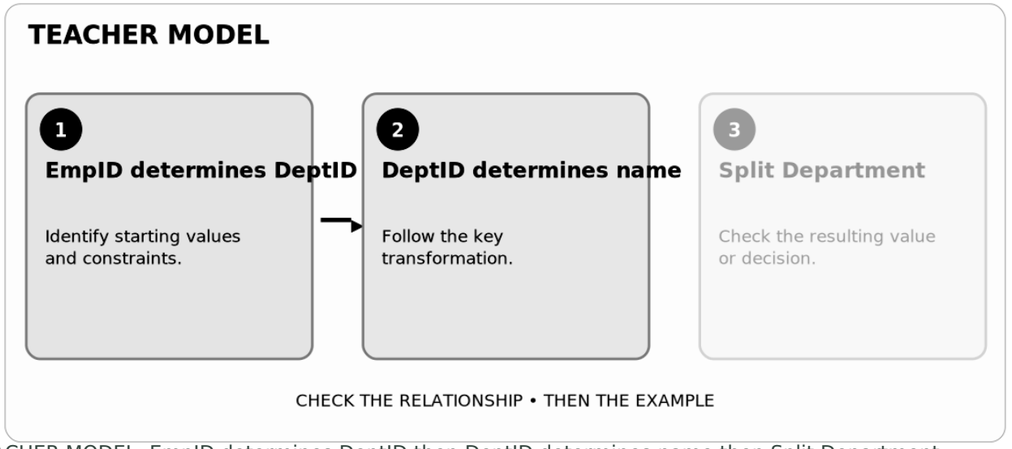
Find the given data and name the required result.




TEACHER MODEL; EmpID determines DeptID then DeptID determines name then Split Department

**Employee(EmpID,EmpName,DeptID,DeptName) is in 2NF. Why may it violate 3NF?**

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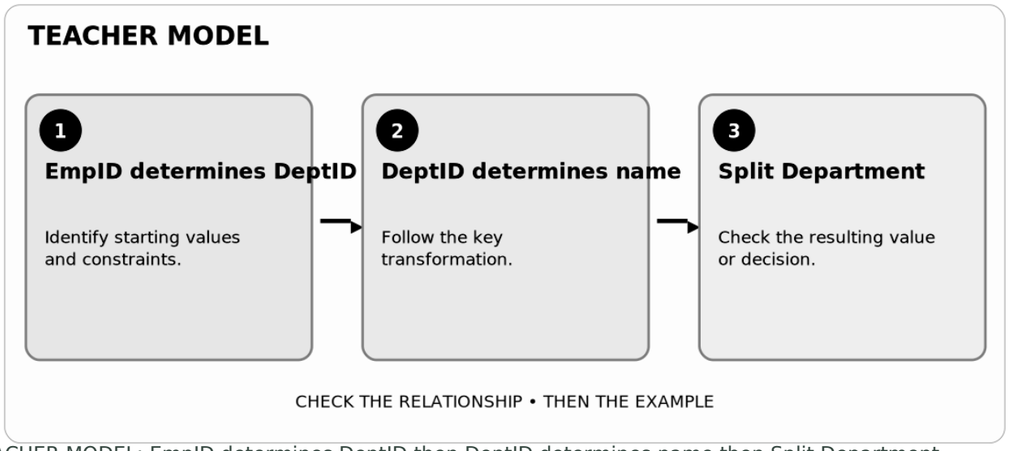

TEACHER MODEL; EmpID determines DeptID then DeptID determines name then Split Department

# I Do · original lesson diagrams

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Find the given data and name the required result.




TEACHER MODEL; EmpID determines DeptID then DeptID determines name then Split Department





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

### Question 3

Explain why normalisation can reduce anomalies but may require additional joins. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

Define transitive dependency and give a practical example. [6]

**Question 2**

Normalise Device(DeviceID,RoomID,RoomName) to 3NF, assuming RoomID determines RoomName. [6]





WE DO — GUIDED PRACTICE

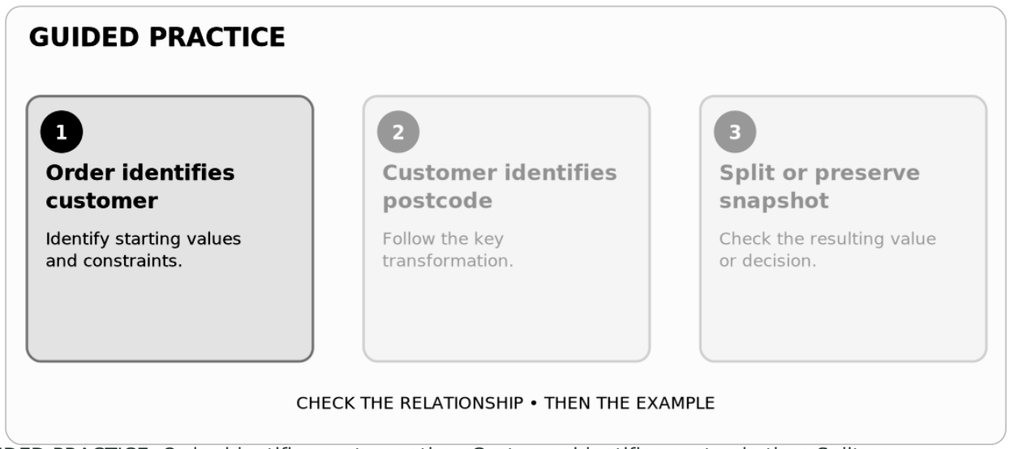
# Try → Discuss → Improve

# We Do · original lesson diagrams

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GUIDED PRACTICE; Order identifies customer then Customer identifies postcode then Split or preserve snapshot





# WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

**Guided 1: Order identifies customer Find the given data and name the required result.**

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.